

H2GUARD · REQUIREMENTS

H2Guard requirements

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DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept estimates
0.1	2026-09-25	Amish Chadha	Initial scaffold

H2Guard requirements

These are first-pass requirements for the concept. Targets are proposals for review, not user-validated needs, and will be checked by calculation at TRL 3 and revised after sessions with the first users. Status is judged against the first-order estimates in HGD-PRC-001. Two requirements are **not met** by the current concept (R12 and R14), and several are met by design only and still unverified.

Table 1. Requirements. LFL is the lower flammability limit of hydrogen in air, taken as 4.0 % by volume, so 10 % LFL is 0.4 % vol and 25 % LFL is 1.0 % vol.

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 2
R1	Measure hydrogen at the high point of the room	0 to 100 % LFL (0 to 4 % vol), resolution 1 % LFL or better; head within 0.3 m of the ceiling above the likely leak source	Sensor datasheet review; installation rule in the precis	Met by selection (catalytic sensor), unverified
R2	Warn early	At 10 % LFL (0.4 % vol): fan to boost, amber light, intermittent sounder; self-clearing	Set point and logic review	Met by design
R3	Trip and shut off the supply	At 25 % LFL (1.0 % vol): valve closed, fan on boost, continuous sounder and beacon; latched until reset with a key and the reading is below 10 % LFL	Set point and logic review	Met by design
R4	Respond fast enough	Sensor t90 of 30 s or less; alarm within 60 s of a step to 1.1 % vol (twice t90, following the ISO 26142 test approach); valve closed within 2 s of the trip	Datasheets, then a bench test at TRL 4	Unverified: catalytic sensor t90 not yet confirmed from a datasheet; estimate about 35 s from leak reaching the head to valve closed
R5	Fail safe	The supply is closed within 2 s of any of: loss of 24 V power, sensor open or short circuit, heater failure, controller watchdog timeout, fan stopped (no tachometer pulses for 10 s)	Failure modes and effects analysis (FMEA)	Met by design (normally closed valve, energize to open); FMEA not yet done
R6	Trip without firmware	A hardware comparator on the catalytic sensor bridge removes valve power at the trip level even if the microcontroller has stopped	Circuit review	Met by design, unverified
R7	Ventilate the room	Continuous exhaust at high level of 5 air changes per hour or more (150 m ³ /h for the 30 m ³ reference room); boost of 10 air changes per hour or more; make-up air at low level on the far side	Fan curve against duct and grille losses	Met on free-air fan rating; unverified with duct losses
R8	Keep the design leak below the trip level	With the design leak of 5 L/min and continuous ventilation, the well-mixed room stays below 10 % LFL	First-order dilution calculation	Met by estimate: about 5 % LFL well mixed; local ceiling values may be several times higher
R9	Limit what can leak	Hydrogen inventory that could be released into the room is 1 % of room volume or less at atmospheric pressure (300 L for 30 m ³), unless a flow restrictor limits release to the design leak	Installation rule and host project review	Rule proposed; applies to the gas system, not to H2Guard hardware
R10	Alert people	Sounder 85 dB(A) or more at 1 m and a red beacon visible from the room entrance	Component ratings	Met by selection (about 90 dB at 1 m), unverified
R11	Keep mains out of the self-built parts	All self-built wiring 24 V DC; mains only inside a certified power supply; power draw 60 W or less	Design review	Met by design (about 21 W normal, about 29 W in alarm, estimates)

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 2
R12	Be checkable in use	Bump test with 1 % vol hydrogen span gas through a test cap in 2 min or less; self-test button runs the alarm, fan boost and valve close; event log of readings, warnings, trips and resets kept for 90 days or more	Design review	Not met: no test cap or gas port is designed yet; log and self-test by design
R13	Install simply	Mounted and wired by one person with hand tools in 4 h or less, excluding the gas fitting, which a competent person makes	Installation walk-through at TRL 4	Unverified
R14	Stay within the concept budget	Parts \$180 or less	Priced BOM	Not met: about \$239 (indicative); budget change proposed, awaiting Amish
R15	Avoid adding ignition sources in the ceiling layer	Fan never switched on in a flammable mixture (runs continuously); controller, relays and power supply mounted at least 1 m below the ceiling; sensor ports behind sintered flame arrestors	Design review	Met by layout; not certified for hazardous areas, and the catalytic element runs hot

■ Assumptions

- Reference room: 4 x 3 x 2.5 m (30 m³), one door, general ventilation unknown and taken as zero.
- Design leak: 5 L/min of hydrogen, representing a failed fitting downstream of a regulator with a flow restrictor. A small electrolyzer of about 100 W makes only about 0.4 L/min (see the precis), so this covers it with margin. A failed regulator on an unrestricted cylinder can release far more, and H2Guard does not protect against that (R9).
- Hydrogen collects first in a layer under the ceiling. The first-order numbers use a 0.3 m layer and treat it as well mixed; real plumes are more concentrated near the source.
- Set points follow common practice for hydrogen detection (warning at 10 % LFL, shutdown at 25 % LFL); ISO 26142 describes alarm levels at 25 % or 50 % LFL. The choice of set points is proposed, awaiting Amish.
- Budget: the \$180 in `project.yaml` covers H2Guard hardware only, not the gas system, span gas or installation labor.